

X1-L1 lab: Capstone 1: refactor the NAT pipeline on your laptop

Platform: Laptop

You need Java 17 or 21 and the `spark` dependency group (see `practice/README.md`). The acceptance tests run on a local `SparkSession` with Delta Lake.

1. Read `practice/capstone/before_pipeline.py` and list every defect you see. Compare your list with the concept section.
2. Read `practice/tests/test_x1_pipeline.py`. Write down, in one line each, what the 11 tests require.
3. Run `pytest practice/tests/test_x1_pipeline.py`. Every test fails with `NotImplementedError`. That is your starting point.
4. Implement `load_config` and `PipelineConfig.post_init`. Run the two config tests until they pass.
5. Implement `ingest_bronze`, then `transform_silver` and `upsert_silver`, then `compute_gold`. Run the tests after each function.
6. Implement `run_checks` and `enforce`, then `build_run_record`, `log_run` and `get_unprocessed_waves`.
7. Run `ruff check practice` and `mypy`. Fix every finding.
8. When all 11 tests pass, compare your module with `practice/solutions/x1_pipeline.py`. Note two things the reference does differently and why.

Notes